Gust Lock – General Data

1. General

Gust locks are provided to secure the main flight control surfaces (ailerons, rudder and elevator) while the aircraft is parked. The rudder system is provided with a built-in system controlled by an operating lever in the flight compartment floor in front of the pilot's seat. Both aileron and elevator gust locks are removable struts (part of flyaway kit) which engage fittings on the instrument panel structure, control column and floor. On aircraft with Mod 6/1676 incorporated, the elevator gust lock strut locks the control column in a forward position. The rudder gust lock must be engaged and the handle held up until the vertical strut is installed. A pivoted flag, Mod 6/1726, provides visual warning that the gust lock is installed. Refer to Figure 1 for details.

A. Rudder Gust Lock

The rudder gust lock is a permanently installed mechanical linkage beneath the flight compartment floor which secures the rudder in the neutral position. The gust lock linkage consists of a hinged lever, connecting rods, a bellcrank and a spring-loaded lever. The hinged lever is flush with the flight compartment floor and when lifted to select rudder gust lock, operates the connecting rods to engage the spring-loaded lever in a slot in the rudder control quadrant. A spring connected to the spring-loaded lever, ensures that the lock cannot be engaged unless selected.

2. Inspection/Check

A. Check Rudder Gust Lock Operation

1) Ensure rudder control system is correctly rigged (refer to 27-20-00, Rudder Control System – General Data).

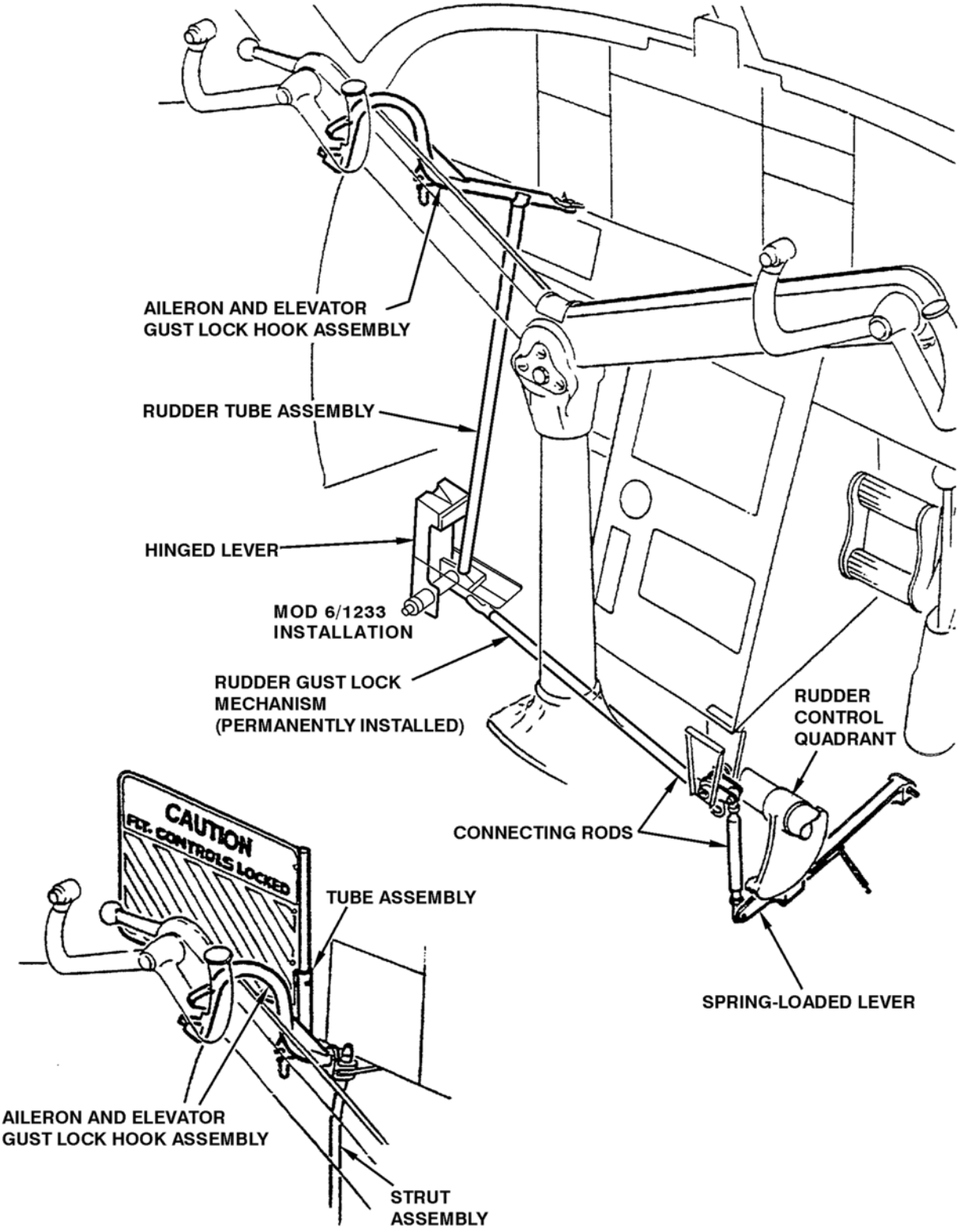
2) Set rudder, aileron and elevator controls to their neutral positions, engage rudder gust lock and install aileron and elevator gust locks.

3) Ensure that the rudder gust lock spring-loaded lever fully engages rudder quadrant.

4) Remove aileron and elevator gust lock struts and lower rudder gust lock hinged lever. Check lever is flush with floor and that spring-loaded lever completely clears rudder control quadrant.

5) On aircraft incorporating Mod 6/1755, check that clearance between plunger in spring-loaded lever and quadrant is 0.120 inch minimum. Adjust connecting rod as necessary and ensure there is no free play between plunger and quadrant when plunger is engaged in gust lock on position.

6) Operate rudder and reengage rudder gust lock several times, checking each time for full and free range of rudder movement after disengaging gust lock.



Gust Lock Installation
Figure 1